Name: Alissa Le
Student ID: 103489136

Introduction

This report aims to evaluate and analyse multiple datasets related to Australian beaches, with the objective of assessing their suitability for further analytical use. The analysis focuses on understanding key characteristics of the data, identifying data quality issues and exploring relationships between important variables such as accessibility, environmental conditions, and visitor numbers.

The process begins with data acquisition and preparation, where each dataset is examined individually to detect inconsistencies, missing values and other issues that may affect analysis. Appropriate data cleaning techniques are then applied to improve data quality and ensure consistency across datasets. Following this, the cleaned datasets are integrated into a single dataset to support more comprehensive analysis.

The second part of the report focuses on data exploration, where both categorical and numerical variables are analysed using descriptive statistics and visualisation techniques. Relationships between variables are further investigated, including the use of correlation analysis and hypothesis testing to provide statistical evidence. Overall, the report aims to generate meaningful insights from the dataset while evaluating its reliability and usefulness for future applications.

Part 1: Data Acquisition & Preparation

1. Data Exploration

There are four datasets in total, including beach_amenities.csv, climate_summary_by_region.csv, data_except_nsw.csv and data_nsw.csv. The process begins with exploring each dataset individually to understand its structure, followed by cleaning each dataset separately before integrating them into a final dataset.

For each dataset, an initial exploration step is carried out to gain a general understanding of the data structure and to document any issues before proceeding with the cleaning process. One of the methods frequently used is pandas DataFrame head() from W3Schools (2024). This method provides a quick view of the column names and sample values, allowing an initial assessment of whether the data appears reasonable. For example, when applying this method to the beach_amenities dataset, it is clear that columns such as BBQ and Has_Showers are stored as strings instead of boolean values, which would need to be corrected for more effective analysis.

Another commonly used method is pandas DataFrame info() along with pandas DataFrame describe(), which provides more detailed information about the dataset. Specifically, the pandas DataFrame info() method displays the number of non-null values and the data type for each column (W3Schools, 2024). This makes it easier to identify mismatches between the actual data types and those specified in the data dictionary. For instance, in both data_except_nsw and data_nsw, the Water_Quality_Rating column is stored as a float instead of an integer, while

Patrolled and Shark_Net are stored as objects instead of boolean values. Similarly, General_Hazard_Rating is stored as a float rather than an integer. These inconsistencies may lead to incorrect analysis if not addressed.

In addition, the pandas DataFrame info() method helps identify missing values through the non-null count. Based on this, a significant number of null values were detected in the Dogs_Allowed and Flags columns in both data_except_nsw and data_nsw.

The pandas DataFrame describe() method provides further insight by summarising statistics such as value counts, number of unique values and maximum values. This is useful for identifying potential duplicate values, unexpected categories and outliers. For example, in the beach_amenities dataset, the Name column contains 301 unique values out of 303 records, suggesting that duplicate entries may exist. Additionally, the Playground column shows four unique values, despite representing a boolean variable, indicating inconsistent data entry.

Outliers were identified by comparing the observed values with the valid ranges specified in the data dictionary. In the climate_summary_by_region dataset, the maximum value of Annual_Rainfall_mm is 2360, which exceeds the allowed range of 200-2000. Similarly, Extreme_Heat_Days has a maximum value of 29, which is outside the valid range of 4-17.

In the data_except_nsw dataset, the State column contains nine unique values, even though only eight states are defined in the glossary, indicating the presence of an invalid value. The Accessibility column also shows nine unique values instead of the expected four, suggesting inconsistent or misspelled entries. Furthermore, the Annual_Visitors column contains values exceeding the maximum allowed range of 5,000,000, indicating potential outliers.

Similarly, in data_nsw, the Name column is not unique and may contain duplicate records. The Accessibility column again shows more unique values than expected, indicating the same issue of inconsistent data entry. The Annual_Visitors column also contains values exceeding the allowed range.

After identifying missing values using the pandas DataFrame info() method, it is necessary to determine whether these values should be handled or removed. To evaluate this, pandas DataFrame isnull() and pandas DataFrame mean() were used together to calculate the percentage of missing values for each column. This approach is effective because isnull() identifies missing entries, while mean() calculates their proportion.

Based on this analysis, the Dogs_Allowed and Flags columns in the data_except_nsw dataset contain 58.5% and 62% missing values, respectively. As these exceed the 50% threshold, these columns are considered unreliable and are therefore suitable candidates for removal.

For the data_nsw dataset, the Dogs_Allowed column contains 62.6% missing values and was removed for the same reason. Although the Flags column contains 43.4% missing values, it was also removed to maintain consistency with data_except_nsw, ensuring both datasets have the same structure for integration.

While the describe() method is useful for identifying maximum values, it does not provide information about minimum values. Therefore, the pandas DataFrame min() method was also used to identify lower-bound outliers. For instance, in the climate_summary_by_region dataset, the minimum value for Extreme_Heat_Days is 0, which is below the valid range of 4–17.

To further investigate outliers, the pandas DataFrame query() method was used to filter values outside the acceptable range. This method allowed for a more detailed examination and confirmed that multiple values fall outside the expected limits.

To confirm the presence of duplicate rows, the pandas DataFrame duplicated() method was used. This method returns a Boolean series indicating whether each row is duplicated. The results show that there are two duplicated rows in both the beach_amenities and data_nsw datasets.

The issue of inconsistent unique values extends beyond duplicate rows and affects categorical columns as well. While the duplicated() method is used to identify duplicate records, the pandas Series unique() method is used to examine the distinct values within each column. This method is particularly useful for identifying inconsistencies such as misspellings and formatting issues.

For example, in the beach_amenities dataset, the Playground column includes values such as “True” instead of “Yes”, as well as values with trailing whitespace, resulting in additional unique categories. In addition, irregular entries were also identified in numerical columns such as Car_Spaces, which includes values like “100?” and “30–50”, indicating inconsistent data formats.

In the data_except_nsw dataset, the State column contains an invalid value “NZ”. The Accessibility column includes misspelled values such as “goood”, “G”, and “Good.”, while the Beach_Type column includes misspellings such as “Lagon” and “Reeef”. Similar issues are also observed in the data_nsw dataset.

2. Data Cleaning

Duplicate rows were identified in both the beach_amenities and data_nsw datasets. The pandas DataFrame drop_duplicates() method was applied to remove redundant records and maintain data integrity. In the beach_amenities dataset, some rows were not entirely identical but shared the same Name with slight differences in other attributes. Since Name is expected to be unique and is used as a key identifier, duplicates were removed based on this column to ensure consistency for later integration.

For values containing whitespace, the pandas Series str.strip() method was used to standardise formatting. This was particularly relevant for columns such as Playground, where leading or trailing spaces caused values to be treated as separate categories.

For incorrect and misspelled values, such as those found in the Accessibility and Car_Spaces columns, the pandas DataFrame replace() method was applied. This allowed invalid entries such as “goood”, “G” or “Good.” to be standardised according to the glossary. The replace()

method was also used to convert invalid or non-standard values into null values where appropriate, making them easier to handle in the next steps.

Data type conversion was carried out using the pandas DataFrame astype() method where possible. This is necessary because many columns were initially stored as object types, which limits their usability in analysis and visualisation. Converting these columns into appropriate data types improves consistency and enables further processing.

However, an issue occurred with the Car_Spaces column. After removing incorrect values, the column contained both numeric values and null values, which resulted in it being stored as a float. Attempting to convert it directly to an integer using astype() produced an error, as integer types do not support null values. To address this, the pandas to_numeric() method with errors = 'coerce' was applied, allowing the conversion to proceed while safely handling invalid entries.

Outliers were handled using the pandas DataFrame where() method, which enables conditional filtering of values. For example, in the Annual_Rainfall_mm column, values outside the acceptable range were replaced. In some cases, this was combined with the pandas Series between() method to retain only values within a defined range. This approach was applied to the Extreme_Heat_Days column, for instance, to ensure that only values between 4 and 17 were kept.

Lastly, the Dogs_Allowed and Flags columns were removed due to their high proportion of missing values (greater than 50%). Retaining these columns would not contribute meaningful information and could negatively affect the analysis. The pandas DataFrame drop() method was used to remove these columns, ensuring consistency across datasets before integration.

3. Integration

After cleaning, all datasets were integrated into a single dataset. Initially, the data_nsw dataset did not contain a State column, unlike the data_except_nsw dataset. If the datasets were combined without addressing this issue, it would introduce a large number of null values in the State column for the data_nsw dataset. To resolve this, the pandas DataFrame insert() method was used to create a State column with the value “NSW” for all rows. This method was chosen because it allows the user to control the exact position of the newly inserted column.

After adding the State column, the pandas concat() method was used to combine the data_nsw and data_except_nsw datasets by stacking them row-wise. This approach is appropriate because both datasets share the same structure and represent similar types of observations. Unlike the merge() method, which combines datasets based on key columns, concat() simply appends one dataset to another, making it suitable for this step. The resulting dataset was named df_aus.

Next, the df_aus dataset was integrated with the beach_amenities dataset using the pandas merge() method. This method combines datasets based on a common key, similar to a SQL join. The datasets were merged using Name as the key identifier, with a left join applied to ensure that all records from df_aus were retained while matching corresponding values from beach_amenities.

Finally, df_aus was integrated with the climate_summary_by_region dataset. Before performing the merge, a mismatch in column names was identified, as the region columns differed between the datasets (Region in df_aus and Region_Name in climate_summary_by_region). To address this, the pandas DataFrame rename() method was used to standardise the column name to Region. The datasets were then merged using the same merge() method with Region as the key identifier and a left join to maintain consistency.

PART B

1. Visualise Categorical Data

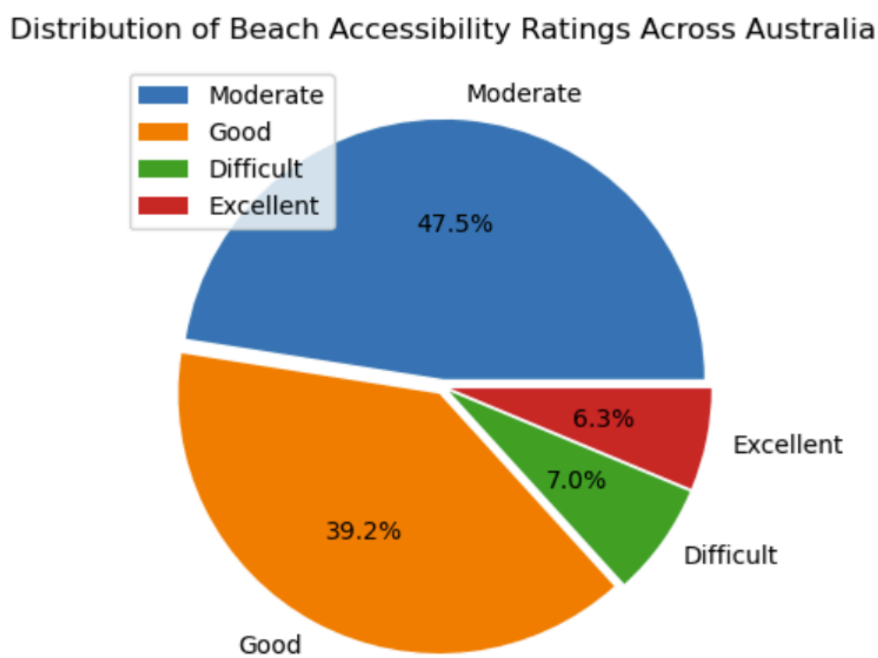

Figure 1: Distribution of Beach Accessibility Ratings Across Australia

For categorical data visualisation, the Accessibility column was selected to examine the distribution of different accessibility ratings across beaches in Australia. This column is relevant as it provides an overall view of how accessible beaches are, which may influence visitor numbers and support further analysis in later sections. For example, identifying whether more beaches fall under “Difficult” or “Excellent” accessibility can help explore potential relationships between accessibility and annual visitor patterns.

Before visualising the data, the pandas Series value_counts() method was applied to calculate the frequency of each accessibility category. This ensures that the distribution is clearly quantified before visualisation.

A pie chart was chosen as it effectively represents the proportion of categorical data. Specifically, it presents each category as part of a whole, making it easier to interpret the relative contribution of each accessibility rating. Since the aim is to compare how each category contributes to the overall dataset, this type of visualisation provides a clear and straightforward comparison.

Based on the chart, most beaches fall under the Moderate (47.5%) and Good (39.2%) accessibility categories. In contrast, only a small proportion are classified as Difficult (7.0%) and Excellent (6.3%).

This result is generally consistent with expectations, as most public beaches are likely designed to be reasonably accessible to a wide range of visitors rather than being highly difficult or fully optimised. However, the relatively low proportion of Excellent accessibility may indicate areas for improvement. Factors such as facility quality or water quality may contribute to this and can be explored further in subsequent analysis.

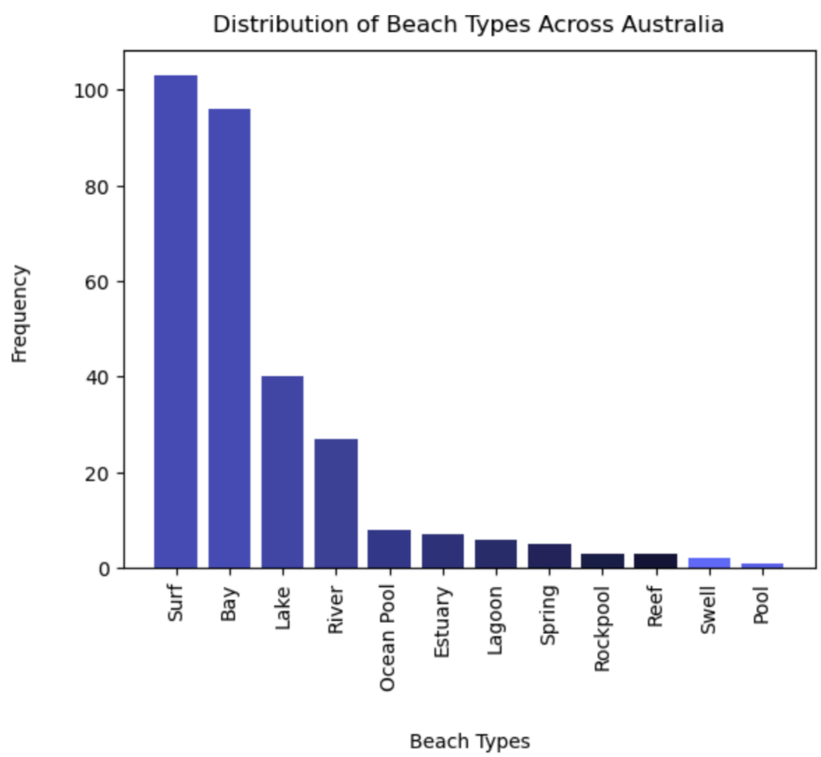

Figure 2: Distribution of Beach Types Across Australia

For another categorical data visualisation, the Beach_Type column was selected to examine the variety and frequency of different beach environments across Australia. This column is useful as it provides context on how different types of beaches are distributed. Initially, it was chosen because it is one of the very few categorical variables that can generate meaningful insights. After reviewing the results, it also became relevant for further analysis, particularly when considering its potential relationship with accessibility. For instance, certain beach types may naturally be more difficult to access than others.

Before visualising the data, the pandas Series value_counts() method was applied to calculate the frequency of each beach type. This ensures that the distribution is clearly summarised and accurately represented in the visualisation.

A bar chart was selected for this analysis as it is more suitable than a pie chart when comparing multiple categories. Specifically, it allows for clear comparison across categories using a common baseline, making differences in frequency easier to interpret. Given that the Beach_Type column contains many distinct values, a bar chart provides a more readable and structured comparison. The use of rotated axis labels further improves clarity due to the number of categories displayed.

From the chart, it can be observed that Surf and Bay are the most common beach types, with significantly higher frequencies than the other categories, followed by Lake as well as River. In contrast, beach types such as Pool, Swell, and Reef appear very infrequently, indicating that although there is a wide range of beach types in Australia, their distribution is not evenly spread across the dataset.

However, the noticeable imbalance between common and rare beach types may affect later analysis, particularly when examining relationships with visitor numbers. Beach types with very few observations may not provide reliable or representative insights.

2. Visualise Numerical Data

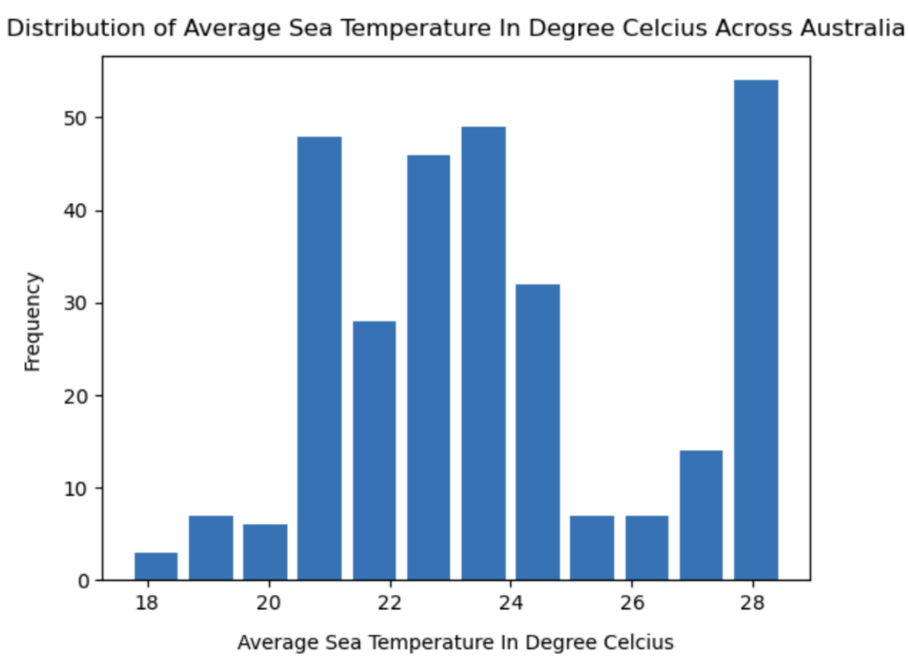

Figure 3: Distribution of Average Sea Temperature In Degree Celcius Across Australia

For numerical data visualisation, the Avg_Sea_Temp_C column was selected to examine how sea temperatures are distributed across beaches in Australia. This variable was chosen because existing research suggests that sea temperature may influence annual rainfall. Therefore, analysing this column helps establish a foundation for exploring a potential relationship between sea temperature and rainfall in later hypothesis testing.

A histogram was selected for this analysis as it is well-suited for continuous numerical data. It groups values into intervals (bins), allowing the overall distribution to be observed more clearly. This makes it easier to identify patterns such as concentration, spread and possible skewness. Compared to other visualisation methods, a histogram provides a straightforward way to understand how frequently values fall within specific ranges.

From the chart, it can be observed that most average sea temperature values are concentrated between approximately 21 Celcius and 25 Celcius, indicating that the majority of beaches fall within a moderate temperature range. There are fewer observations at the lower end (18 Celcius) and at the higher end (28 Celcius).

This distribution suggests that Australia's coastal regions generally experience stable and moderate sea temperatures. However, the presence of some higher temperature values may indicate specific regions with warmer climates, which could be useful for further analysis when examining relationships with other variables such as annual rainfall.

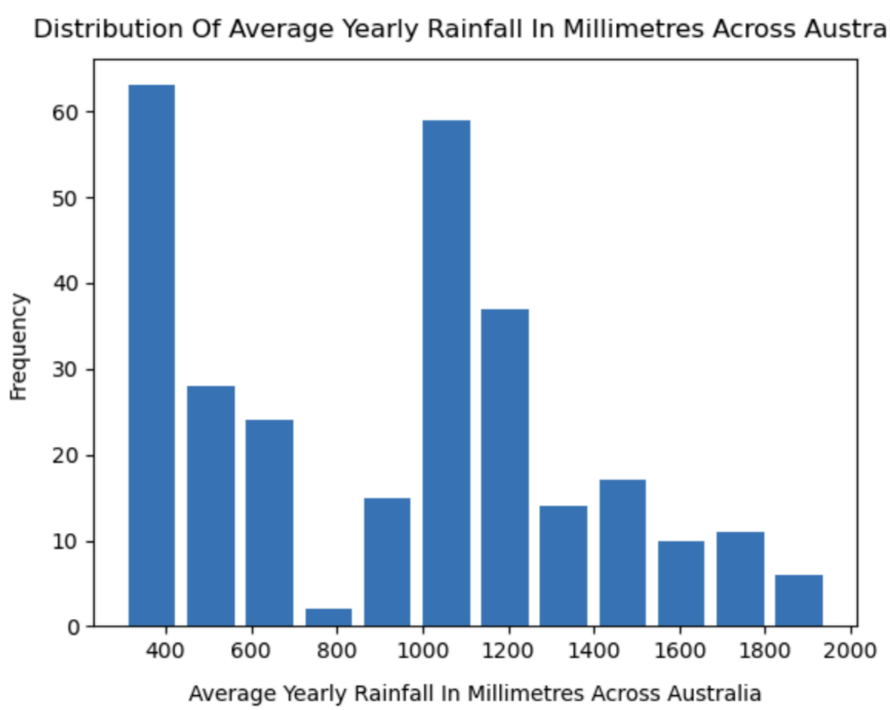

Figure 4: Average Yearly Rainfall In Millimetres Across Australia

For numerical data visualisation, the Annual_Rainfall_mm column was selected to examine how rainfall levels are distributed across different regions in Australia. This variable is particularly relevant as it represents an important environmental factor that may influence beach conditions and, potentially, visitor patterns. In addition, it is closely related to the previously analysed Avg_Sea_Temp_C variable, making it suitable for exploring a possible relationship between sea temperature and rainfall in later analysis.

A histogram was chosen for this analysis as it is appropriate for continuous numerical data. It groups values into intervals (bins), similar to the approach used for average sea temperature, allowing the distribution of rainfall levels to be observed in terms of frequency.

The chart shows that rainfall values are spread across a relatively wide range, approximately from 300 mm to 1900 mm. A noticeable concentration appears in the mid-range (around 900 mm to 1200 mm), suggesting that many regions experience moderate rainfall levels. At the same time, there are still observations at both the lower and higher ends, indicating variation in rainfall across different regions.

This variation provides a useful basis for further analysis, particularly when examining potential relationships with other variables such as sea temperature or annual visitors.

3. Relationship Between Columns

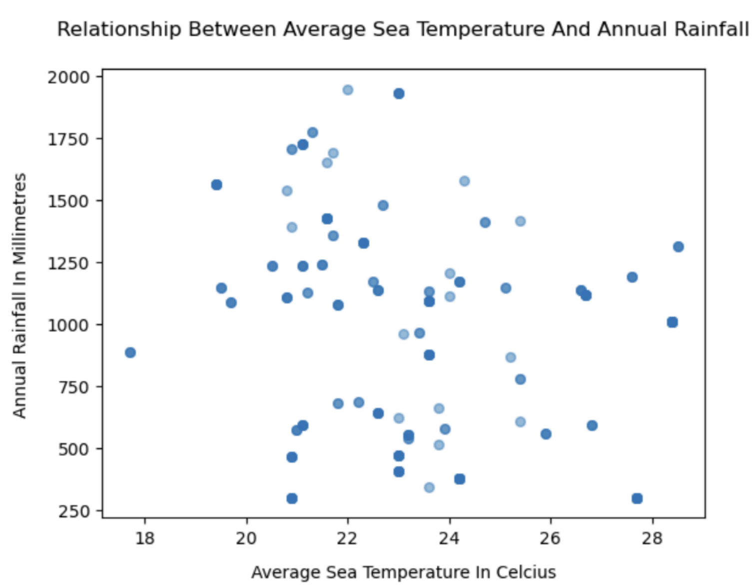

Figure 5: Relationship Between Average Sea Temperature And Annual Rainfall

To explore the relationship between variables, Avg_Sea_Temp_C and Annual_Rainfall_mm were selected. These two variables were chosen because they both represent environmental conditions, and there is a reasonable assumption that sea temperature may be associated with rainfall patterns. This makes them suitable for forming a plausible hypothesis and further statistical testing.

Before analysing the relationship, descriptive statistics were reviewed using the pandas DataFrame describe() method. From the summary, the average sea temperature is approximately 23.73 Celcius, while the average annual rainfall is around 912.87 mm. The relatively small standard deviation for sea temperature compared to rainfall suggests that temperature is more stable, whereas rainfall varies more significantly across regions. This difference in variability may affect how strongly the two variables are related.

A scatter plot was chosen to visualise the relationship between these two numerical variables. This type of visualisation is appropriate as it allows patterns, trends, or correlations between two continuous variables to be observed directly. Each point represents a beach location, making it possible to assess whether higher temperatures are associated with higher or lower rainfall.

From the scatter plot, there is no clear linear pattern between average sea temperature and annual rainfall. The data points are widely dispersed, indicating that rainfall does not consistently increase or decrease with temperature. While some higher rainfall values appear around mid-range temperatures, the pattern is not strong enough to suggest a clear relationship.

To support this observation, the correlation coefficient was calculated. The result shows a weak negative correlation (approximately - 0.23), indicating a slight tendency for rainfall to decrease as temperature increases. However, this relationship is relatively weak and does not suggest a strong association between the two variables.

This result does not strongly support the initial assumption that sea temperature significantly influences rainfall. While there is a slight negative trend, it is not substantial enough to conclude a meaningful relationship. Therefore, other factors may play a more important role in determining rainfall patterns.

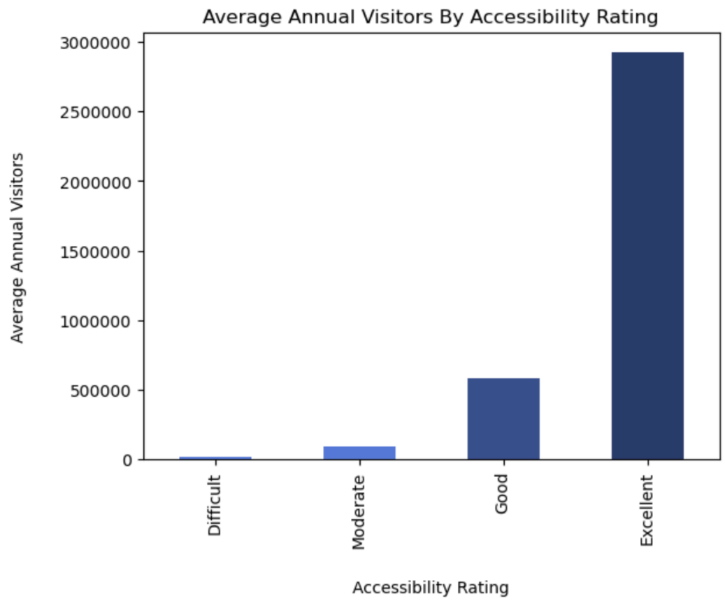

Figure 6: Average Annual Visitors By Accessibility Rating

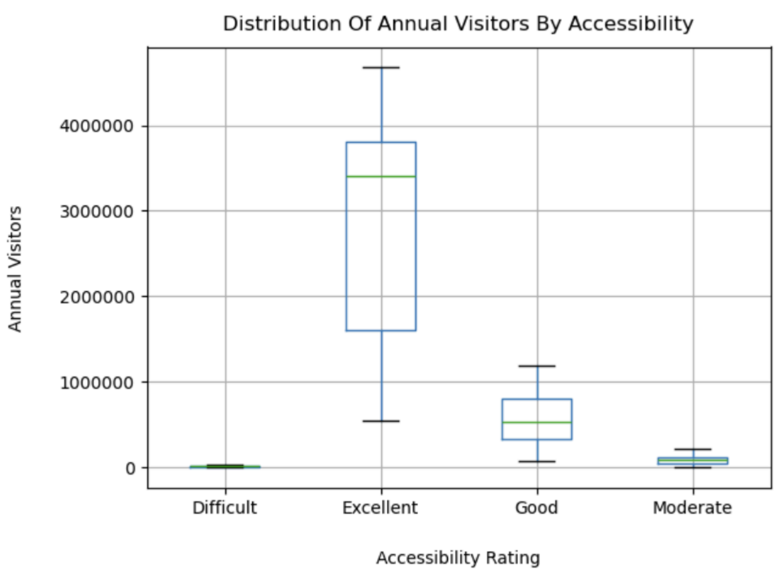

Figure 7: Distribution of Annual Visitors By Accessibility

To further explore relationships within the dataset, the Accessibility and Annual_Visitors columns were selected. Accessibility is an important factor that may influence how easily visitors can reach a beach, while annual visitors represent the level of usage. Therefore, examining these two variables together helps to understand whether accessibility plays a role in attracting visitors.

To begin the analysis, the average number of annual visitors was calculated for each accessibility category using the pandas DataFrame groupby() method combined with mean().

This aggregation is necessary as it summarises the overall visitor trend for each category, making comparisons more meaningful.

A bar chart was used to visualise the average annual visitors across accessibility ratings. This type of visualisation is suitable as it allows a clear comparison between discrete categories. The results show a clear increasing trend: beaches with Excellent accessibility have the highest average number of visitors, followed by Good, Moderate and Difficult. This suggests a positive relationship between accessibility and visitor numbers.

To further examine this relationship, the categorical accessibility ratings were converted into numerical values (from 1 to 4) to allow correlation analysis. Using np.corrcoef(), the correlation coefficient was calculated to be approximately 0.69, indicating a moderately strong positive relationship between accessibility and annual visitors. This supports the pattern observed in the bar chart.

In addition, a box plot was used to visualise the distribution of annual visitors within each accessibility category. This provides a more detailed view compared to the average alone, as it shows the spread, median and potential outliers. The box plot confirms that beaches with higher accessibility ratings tend to have higher visitor numbers overall. It also highlights variability within categories, particularly for Excellent and Good, where the range of visitor numbers is wider.

4. Hypothesis

To formally test whether accessibility has an influence on annual visitors, a hypothesis test was conducted using one-way ANOVA. This method is appropriate as it compares the mean annual visitors across more than two groups, in this case, the four accessibility categories (Difficult, Moderate, Good and Excellent).

The hypotheses are defined as follows:

- Null Hypothesis: Accessibility rating does not influence annual visitors (the mean number of visitors is the same across all categories).
- Alternative Hypothesis: Accessibility rating influences annual visitors (at least one category has a different mean).

To perform the test, the annual visitor values were separated into four groups based on accessibility rating. The scipy.stats.f_oneway() method was then applied to compare the group means.

The ANOVA result produced an F-statistic and a p-value. Since the p-value is significantly smaller than the chosen significance level of 0.05, the null hypothesis is rejected.

This result provides strong statistical evidence that there are significant differences in the mean number of annual visitors across different accessibility ratings. In other words, accessibility appears to have a meaningful influence on visitor numbers.

This finding is consistent with the earlier exploratory analysis, where both the bar chart and correlation analysis suggested a positive relationship between accessibility and annual visitors. Therefore, the statistical test reinforces the conclusion that beaches with higher accessibility tend to attract more visitors.

5. Filter Subnet

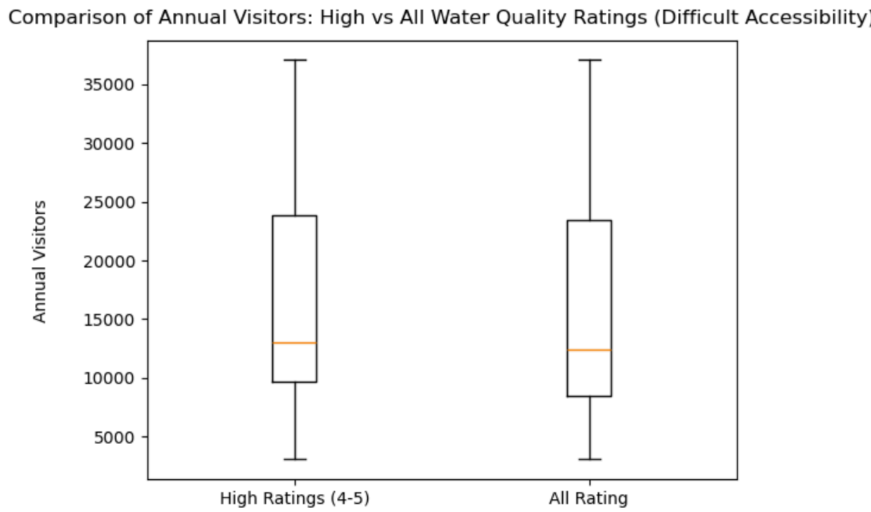

Figure 8: Comparison of Annual Visitors: High vs All Water Quality Ratings (Difficult Accessibility)

To further investigate the factors influencing visitor numbers, a subset of the data was analysed, focusing on beaches with Difficult accessibility. This subset was selected because earlier results showed that beaches with lower accessibility generally attract fewer visitors. Therefore, it is meaningful to examine whether other factors, such as Facilities_Rating and Water_Quality_Rating, can help improve visitor numbers even when accessibility is limited.

The data was first filtered to include only beaches with Difficult accessibility. From this subset, an additional filter was applied to identify beaches with higher ratings (values of 4 or 5) in either Facilities_Rating or Water_Quality_Rating. This allows comparison between higher-quality beaches and the overall group within the same accessibility level.

Descriptive statistics were used to calculate the mean number of annual visitors for both groups. The results show that beaches with higher facility or water quality ratings have an average of approximately 16,089 visitors, compared to 15,367 visitors for all beaches with difficult accessibility. While the difference is not large, it suggests a slight increase in visitor numbers when quality-related factors are higher.

A box plot was used to visualise the distribution of annual visitors between the two groups. This type of visualisation is appropriate as it shows not only the median but also the spread and variability of the data. From the chart, both groups show a similar distribution, with overlapping ranges and comparable medians. This indicates that improving facilities or water quality alone may not lead to a substantial increase in visitors for beaches that are difficult to access.

Conclusion And Future Work

This report has demonstrated a structured approach to analysing multiple beach-related datasets, starting from data acquisition and preparation through to data exploration and statistical testing. Key data quality issues were identified, including missing values, inconsistent data types, duplicate records and outliers. These issues were addressed using appropriate data cleaning techniques, resulting in a more reliable and consistent dataset for analysis.

The exploratory analysis provided several important insights. Most beaches in Australia fall under moderate and good accessibility categories, while excellent accessibility remains relatively limited. Environmental variables such as sea temperature were found to be relatively stable, whereas rainfall showed greater variability across regions. The relationship analysis between sea temperature and rainfall indicated only a weak association, suggesting that rainfall is influenced by additional factors not captured in the dataset.

In contrast, accessibility was found to have a stronger relationship with annual visitors. Both visualisation and correlation analysis indicated a positive trend, which was further confirmed through hypothesis testing using ANOVA. The results showed that accessibility has a statistically significant influence on visitor numbers, with higher accessibility associated with higher visitor counts. Additional subset analysis suggested that while facilities and water quality may slightly improve visitor numbers, their impact is relatively limited compared to accessibility.

Despite these findings, there are several areas for future work. Further analysis could include post-hoc testing following ANOVA to identify which specific accessibility groups differ significantly. Additional variables, such as geographical location or seasonal factors, could also be incorporated to better understand visitor behaviour. Moreover, advanced modelling techniques, such as regression analysis, could be applied to quantify the combined effect of multiple variables on annual visitors.

Overall, while the dataset provides valuable insights, further refinement and extended analysis would enhance its usefulness for decision-making and practical applications.

Acknowledgement

No Generative AI tools were used for this task. I was trying to do it by myself since I want to learn and remember faster by doing. If I get stuck, I would search the lecture slide or W3Schools or YouTube.

References

W3Schools. (2024). *Pandas Tutorial*. Www.w3schools.com. https://www.w3schools.com/python/pandas/default.asp